Michael Zang

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EDUCATION

University of Southern California | Los Angeles, CA — Aug 2026 – May 2028 (Expected)
M.S. Computer Science (Incoming)

University of Notre Dame | Notre Dame, IN — Aug 2022 – May 2026
B.S. Computer Science, Minor in Business Economics – Sheedy Family Scholar
Coursework: Data Science, AI, Database, Algorithms, Data Structures, Linear Algebra, Calculus, and Statistics

PROFESSIONAL EXPERIENCE

University of Notre Dame – CSE Department | Notre Dame, IN — Aug 2024 - Present
Computer Vision Research under Dr. Kevin W. Bowyer (IEEE/IAPR/AAAS Fellow) | First Author (arXiv, WACV '27)
- Curated a 2,000-image in-the-wild monozygotic twin dataset and benchmarked ArcFace, AdaFace, UniFace, and MagFace using 10-fold cross-validation, achieving 76.50% verification accuracy.
- Retrained an ArcFace model without flip augmentation and applied RISE saliency analysis to investigate twin distinguishability, identifying mirror twins and skin marks as key features.

Carnival Cruise Line | Miami, FL — May 2025 – July 2025
Data Scientist Intern - Fuel Performance & Decarbonization
- Engineered a regression-based optimization tool in Python with a custom GUI, reverse engineering Eniram's logic to model ship fuel consumption. Scoped for fleet-wide deployment across 29 ships.
- Built data pipeline (MMDetection, OCSort) with automated minute-level scheduling to process live CCTV feeds across 50+ cameras, storing results in Amazon Neptune for HVAC efficiency and data visualization.

Lenovo Group Limited | Beijing, China — June 2024 – Aug 2024
Data Scientist Intern – Lenovo Research AI Lab
- Leveraged Apriori, decision trees, and LLMs to uncover hidden stock price relationships among major competitors, achieving a 79.42% confidence rate between Lenovo, SMIC, Digital China, and Sunny.
- Queried a 30-million-row SQL database, integrated NLP-derived signals and prior machine learning model through feature engineering a one-step stock prediction model, achieving R^2 scores up to 0.81.

SELECTED PROJECTS

Steam Game Lifecycle Analysis (Full-stack) | Richmond Hill, ON | [Video Game Lifecycle]() — Aug 2026 – Present
- Analyzing 6,000+ Steam games to identify player retention patterns using SQL and Pandas. Applying clustering and survival analysis to model game lifecycles and predict long-term player trajectories.

Car Plate Prediction Website (Full-stack) | Richmond Hill, ON | [Car Plate Recognition]() — May 2026 – Aug 2026
- Developed a YOLO and Tesseract license plate recognition system on 4,500 images, achieving 78.0% mAP@50–95. Built a Dockerized FastAPI/React website for bounding box visualization and analysis.

Closet Recommendation Website (Project Lead) | Notre Dame, IN | [Loverboy Closet]() — Sep 2024 – Dec 2026
- Led research and mentored on recommender systems in PyTorch (matrix factorization, NCF, LightGCN).
- Built back-end for outfit recommendation and subscription, integrating JWT authorization with a MySQL relational database. Cleaned a 30,000-item dataset by engineering a multi-stage regex parsing pipeline.

LEADERSHIP AND ACTIVITIES

Student International Business Club (Amazon) | Notre Dame, IN — Feb 2022 – May 2024
- Designed chatbot for Amazon Health, addressing mental health challenges by connecting users directly with professionals, reducing fees, and fostering trust. Presented to executives in the Boston office.

CS For Good | Notre Dame, IN — Sep 2023 – May 2024
- Led the Dismas House project, focused on reintegration of former prison members into society.
- Modernized organization's website, updated residents ID database, and conducted technology classes.

HIGHLIGHTED SKILLS

Languages: Python, SQL, C **Data:** Pandas, NumPy, Scikit-learn, Spark, AWS **ML/CV:** OpenCV, PyTorch, YOLO, Recommender Systems **Tools:** Docker, Git, React, Flask, Linux **Analytics:** A/B Testing, Power BI, Tableau, Excel